M1.3 — Bucket 2 claims

Live working Markdown for Bucket 2 claim wording, argument, and probe/control evidence. Claims here need new fixtures, controls, or interventions beyond the locked M12x Stage-3 package. Refine one at a time; none are locked until their write-up is complete. TeX companion: `docs/report/findings/milestone1_part3_bucket2_claims.tex`.

1 Claim 1 — Causal-role underdetermination under learner-input equivalence

Claim. Let $\mathcal{T} = (\mathcal{F}_{\mathrm{BK}}, E^+, E^-, C)$ denote the learner-visible task for a fixed configuration C. If two causal designs (G, t) and (G', t) induce the same $\mathcal{T}$, but assign the direct-parent role to different extensionally equivalent predictors, unguided ABA Learning has no learner-visible basis for preferring either predictor on causal grounds. The formal learning problem is the same under both graphs, and the implemented search receives the same input. Any preference between the predictors must therefore arise from representation or search bias rather than from evidence about their causal roles. Consequently, the same fully covering framework can be mechanism-aligned under one graph and misaligned under the other.

Argument. An ABA Learning task is specified by an initial ABA framework, positive and negative examples, and a reasoning mode. Its solution condition concerns acceptance of E^+ and rejection of E^-. It does not refer to G, $\mathrm{Pa}(t)$, or the distinction between a parent and an extensionally equivalent non-parent. The current tabular pipeline preserves this separation: it serialises non-target columns into exact-value BK, while handcrafted graph edges are used only for graph-relative evaluation.

M13-C1 demonstrates the consequence using the shared table

$$\mathcal{D} = \{(x_0, x_1, x_2) = (0, 0, 0), (1, 1, 1), (2, 2, 2)\},$$

with target x_2, $E^+ = \{x_2(2), x_2(3)\}$, and $E^- = \{x_2(1)\}$. Under G_0, x_0 is the common cause of x_1 and x_2, so $\mathrm{Pa}_{G_0}(x_2) = \{x_0\}$. Under G_1, x_1 is the common cause of x_0 and x_2, so $\mathrm{Pa}_{G_1}(x_2) = \{x_1\}$. The ordered table, BK, target, examples, and configuration are identical. Stage-0 tests and the generated comparison confirmed byte-identical data and BK, identical examples, identical normalised learned δ, and identical ASP coverage within both configuration pairs.

Under ECAI, both graph-labelled cells learned

$$x_2(A) \leftarrow x0\_val\_1(A), \qquad x_2(A) \leftarrow x0\_val\_2(A).$$

The traces show one-token non-deterministic folding selecting the first matching x_0 predicates; no assumptions or contraries were introduced. These rules are an exact parent-based solution under G_0, but use only the extensionally equivalent sibling under G_1. Nevertheless, both cells cover all 2/2 positives and reject the 1/1 negative. Thus the same learned framework changes from `clean_recovery=1` to `clean_recovery=0` solely because the external graph-relative interpretation changes.

Under AAMAS, both cells learned

$$x_2(A) \leftarrow x0\_val\_1(A), x1\_val\_1(A),$$
$$x_2(A) \leftarrow x0\_val\_2(A), x1\_val\_2(A).$$

Greedy folding accumulated both matching predictors for each positive row. The learned framework therefore contains the parent and the extensionally equivalent sibling under either graph, with variable-level parent precision 1/2, recall 1, and full ASP coverage. It expresses no preference between their causal roles.

The new paired control isolates what the M12x results could only illustrate. Bucket 1 showed that ECAI's x_0-keyed rules and AAMAS's full-row conjunctions can cite non-parents while retaining sample separation. M13-C1 holds the complete learner-visible task fixed and changes only the valid causal interpretation. The resulting graph-invariant learned frameworks confirm that coverage cannot resolve causal role when the relevant distinction is absent from the learning input. This is an information limit of the unguided task; it does not imply that ABA Learning cannot recover a true parent when the available representation and search happen to select it.

Evidence. Experiment record: `docs/experiments/qualitative/M13-C1-causal-role-underdetermination/`
Generated comparison: `causal/outputs/aba_learning/grid/M13_c1_role_equivalence_summary.md`.